Light

GCSE Computer Science — Character Sets (Set 3)

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GCSE • Edexcel • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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Multiple-choice topic test for **GCSE Computer Science (Edexcel)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

A keyboard has over 100 keys including letters, numbers, and symbols. What does a character set define?

- **A.** The physical layout of keys on a keyboard
- **B.** The unique binary code assigned to each character
- **C.** The font style used to display text on screen
- **D.** The speed at which characters can be typed

Q2 (1 mark)

Standard ASCII is described as a 7-bit character encoding. What does '7-bit' mean in this context?

- ○ **A.** ASCII was invented in 1977
- ○ **B.** Each character is represented using 7 binary digits
- ○ **C.** ASCII can only encode 7 different characters
- ○ **D.** ASCII files must be exactly 7 bytes in size

Q3 (1 mark)

In ASCII, uppercase letters are assigned codes starting from 65 for 'A'.
What is the ASCII code for the letter 'G'?

- ○ **A.** 70
- ○ **B.** 71
- ○ **C.** 72
- ○ **D.** 103

Q4 (1 mark)

In ASCII, lowercase letters begin at code 97 for 'a'.
What is the ASCII code for the lowercase letter 'd'?

- ○ **A.** 68
- ○ **B.** 97
- ○ **C.** 100
- ○ **D.** 132

Q5 (1 mark)

In ASCII, digit characters start at code 48 for '0'.
What is the ASCII code for the digit character '5'?

- ○ **A.** 5
- ○ **B.** 50
- ○ **C.** 53
- ○ **D.** 101

Q6 (1 mark)

A student says: 'Standard ASCII uses 7 bits, so it can represent 14 characters because 7 × 2 = 14.'
Why is this statement incorrect?

- ○ **A.** The correct calculation is 7 + 2 = 9 characters
- ○ **B.** The correct calculation is 2^7 = 128 characters
- ○ **C.** ASCII actually uses 8 bits, not 7 bits
- ○ **D.** The number of characters depends on the computer model

Q7 (1 mark)

In ASCII, uppercase 'P' has code 80.
What is the ASCII code for lowercase 'p'?

- ○ **A.** 48
- ○ **B.** 80

- C. 112
- D. 160

Q8 (1 mark)

The ASCII code for the letter 'R' is 82.
What is 82 in 7-bit binary form?

- A. 1010010
- B. 1010011
- C. 1010100
- D. 1100010

Q9 (1 mark)

Standard ASCII assigns codes from 0 to 127.
Which range contains the uppercase letters A to Z?

- A. 0 to 31
- B. 48 to 57
- C. 65 to 90
- D. 97 to 122

Q10 (1 mark)

A website needs to display content in Hindi, which uses the Devanagari script.
Why can standard ASCII not be used for this website?

- ○ **A.** ASCII is too slow for modern websites
- ○ **B.** ASCII only includes characters for English text
- ○ **C.** ASCII files cannot be transmitted over the internet
- ○ **D.** ASCII requires special hardware to display

Q11 (1 mark)

A short message contains 200 characters.
If stored using 7-bit ASCII, how many bits are needed?

- ○ **A.** 200 bits
- ○ **B.** 207 bits
- ○ **C.** 1,400 bits
- ○ **D.** 1,600 bits

Q12 (1 mark)

Extended ASCII uses 8 bits instead of 7 bits per character.
How many additional characters does extended ASCII provide compared to standard ASCII?

- ○ **A.** 1 additional character
- ○ **B.** 8 additional characters
- ○ **C.** 64 additional characters
- ○ **D.** 128 additional characters

Q13 (2 marks)

A programmer stores the word 'TYPE' using ASCII.
The ASCII codes are: T=84, Y=89, P=80, E=69.
What is the sum of all ASCII codes for this word?

- ○ **A.** 298
- ○ **B.** 310
- ○ **C.** 322
- ○ **D.** 354

Q14 (2 marks)

A text editor converts lowercase to uppercase by subtracting 32 from ASCII codes.
The lowercase letter 'v' has ASCII code 118.
What uppercase letter results from this conversion?

- ○ **A.** T (code 84)
- ○ **B.** U (code 85)
- ○ **C.** V (code 86)
- ○ **D.** W (code 87)

Q15 (2 marks)

Two versions of a document are saved:
• Version A: 4,000 characters in 7-bit ASCII
• Version B: 3,200 characters in 8-bit extended ASCII
Which version requires more storage?

- ○ **A.** Version A (28,000 bits)

- ○ **B.** Version B (25,600 bits)
- ○ **C.** Both versions are exactly the same size
- ○ **D.** Version B (32,000 bits)

Q16 (2 marks)

A computer receives the 7-bit binary code 1001000.
Using ASCII, what character does this represent?

- ○ **A.** F (code 70)
- ○ **B.** G (code 71)
- ○ **C.** H (code 72)
- ○ **D.** I (code 73)

Q17 (2 marks)

A games company creates a character set with exactly 100 unique symbols for their game.
What is the minimum number of bits needed to represent all 100 symbols?

- ○ **A.** 6 bits (64 values)
- ○ **B.** 7 bits (128 values)
- ○ **C.** 8 bits (256 values)
- ○ **D.** 100 bits (100 values)

Q18 (2 marks)

A program sorts strings by comparing ASCII codes character by character from left to right.
Given: 'Apple' and 'Banana'.
In ASCII: A=65, B=66.
Which string comes first in ascending order?

- ○ **A.** 'Banana' because it has more letters
- ○ **B.** 'Apple' because A (65) is less than B (66)
- ○ **C.** 'Banana' because B (66) is greater than A (65)
- ○ **D.** 'Apple' because it has fewer letters

Q19 (2 marks)

Unicode was developed to address limitations of ASCII.
What is the main advantage of Unicode over ASCII?

- ○ **A.** Unicode files are smaller than ASCII files
- ○ **B.** Unicode can process text faster than ASCII
- ○ **C.** Unicode can represent characters from virtually all world writing systems
- ○ **D.** Unicode requires less memory to store each character

Q20 (2 marks)

A text file contains 'LINE1' followed by a carriage return (code 13), line feed (code 10), then 'LINE2'.
How many characters are stored in total?

- ○ **A.** 10 characters

- B. 11 characters
- C. 12 characters
- D. 23 characters

Q21 (2 marks)

An email message contains 2,800 characters stored in 8-bit ASCII.
What is the file size in bytes?
(1 byte = 8 bits)

- A. 350 bytes
- B. 2,800 bytes
- C. 19,600 bytes
- D. 22,400 bytes

Q22 (2 marks)

A password 'Gd5!' contains an uppercase letter, lowercase letter, digit, and symbol.
Using ASCII: G=71, d=100, '5'=53, '!'=33.
How many bits are needed to store this password in 7-bit ASCII?

- A. 4 bits
- B. 28 bits
- C. 32 bits
- D. 257 bits

Q23 (2 marks)

Extended ASCII keeps codes 0-127 identical to standard ASCII, then adds new characters in codes 128-255.
Why was this design choice important?

- ○ **A.** It made extended ASCII files exactly twice the size of standard ASCII
- ○ **B.** It ensured standard ASCII documents would display correctly on extended ASCII systems
- ○ **C.** It allowed extended ASCII to use fewer bits per character
- ○ **D.** It made extended ASCII faster to process

Q24 (3 marks)

A telecommunications company designs a character set for text messaging that must include:
• 26 uppercase letters
• 26 lowercase letters
• 10 digits
• 15 punctuation marks
• 10 special symbols (like emojis)
What is the minimum number of bits needed per character?

- ○ **A.** 6 bits (64 possible values)
- ○ **B.** 7 bits (128 possible values)
- ○ **C.** 8 bits (256 possible values)
- ○ **D.** 87 bits (one per character type)

Q25 (3 marks)

A library digitises book catalogues. Each record averages 250 characters.
Compare two encoding options for 80,000 records:
• Option A: 7-bit ASCII
• Option B: 8-bit extended ASCII
How much additional storage does Option B require?

- ○ **A.** 20,000,000 bits (2,500,000 bytes)
- ○ **B.** 80,000 bits (10,000 bytes)
- ○ **C.** 250,000 bits (31,250 bytes)
- ○ **D.** 140,000,000 bits (17,500,000 bytes)

Q26 (3 marks)

A student makes these statements about ASCII:
1. ASCII code 65 represents 'A' on every computer that uses ASCII
2. Lowercase letters have higher ASCII codes than uppercase letters
3. The digit character '9' has a higher ASCII code than the digit character '1'
4. ASCII can represent all punctuation marks used in every language
Which statements are correct?

- ○ **A.** Statements 1 and 2 only
- ○ **B.** Statements 1, 2, and 3 only
- ○ **C.** Statements 2, 3, and 4 only
- ○ **D.** All statements are correct

Q27 (3 marks)

A programmer receives a file where the word 'café' displays as 'cafÃ©' on their system.
The file was created on a different computer.

What is the most likely cause of this display problem?

- ○ **A.** The file has been corrupted during transfer and cannot be recovered
- ○ **B.** The sending computer used a different encoding than the receiving computer expects
- ○ **C.** The letter 'é' is impossible to store in any computer system
- ○ **D.** The programmer's monitor is faulty and needs replacement

Q28 (3 marks)

A validation algorithm checks if a character is a digit:
`IF ASCII code >= 48 AND ASCII code <= 57:`

```
  Return TRUE (is a digit)
ELSE:
  Return FALSE (not a digit)
```

For which input will the algorithm return TRUE?

- ○ **A.** The letter 'D' (ASCII 68)
- ○ **B.** The digit character '0' (ASCII 48)
- ○ **C.** The number fifty (no ASCII code)
- ○ **D.** The letter 'a' (ASCII 97)

Q29 (3 marks)

A sensor transmits temperature readings as text over a noisy communication link. The reading '72' is sent but one bit in the first character is corrupted.
Original '7' (ASCII 55, binary 0110111) becomes 0100111 (ASCII 39).

What will the receiver display?

- ○ **A.** An error message because the data is invalid
- ○ **B.** The reading '72' because error correction fixes it automatically
- ○ **C.** The characters "'2" (apostrophe followed by 2)
- ○ **D.** A blank display because sensors cannot send corrupted data

Q30 (3 marks)

A database stores 500,000 product codes. Each code is exactly 8 characters using only uppercase letters (A-Z) and digits (0-9).
A developer proposes a custom 6-bit encoding instead of 8-bit ASCII.
How much storage would be saved in total?

- ○ **A.** 500,000 bits (62,500 bytes)
- ○ **B.** 1,000,000 bits (125,000 bytes)
- ○ **C.** 4,000,000 bits (500,000 bytes)
- ○ **D.** 8,000,000 bits (1,000,000 bytes)

Total: **55** marks • Time: **30 min**

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